

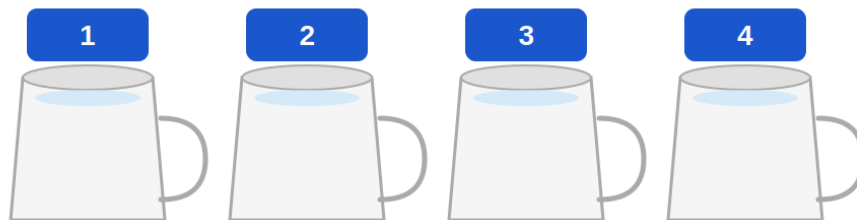
Tetrad Sensory Evaluation Test

Incorrect Cup Labeling — Common Mistakes

A tetrad test presents panelists with four cups: two containing Sample A and two containing Sample B. The panelist must identify which two cups are alike. For valid results, all four cups should be labeled with unique, randomly generated 3-digit codes that give no information about sample identity, position, or grouping.

The four errors below are common labeling mistakes that compromise test validity.

Error 1: Sequential numbers



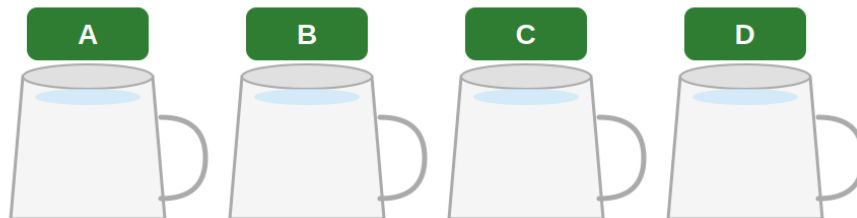
Why it's wrong

Numbering cups 1, 2, 3, 4 reveals the presentation order and implies a ranking or grouping relationship between cups.

Bias introduced

Panelists infer that cups 1 & 2 and cups 3 & 4 are pairs, or that adjacent numbers share a sample. Positional bias is introduced.

Error 2: Alphabetical / letter labels



Why it's wrong

Letters such as A, B, C, D are standard shorthand in sensory science for the two products being compared. Using them as labels may lead panelists to assume samples are different, whether or not they are.

Bias introduced

Panelists recognize 'A' and 'B' as the two sample types. The test is no longer blind, and confirmation bias skews responses.

Error 3: Identical codes on matched pairs



Why it's wrong

Assigning the same 3-digit code to both cups of the same sample (e.g., two cups labeled '537' and two labeled '291') reveals which cups are paired.

Bias introduced

The panelist trivially identifies the pairs by matching codes, making the discrimination task meaningless. All four cups must carry unique codes.

Error 4: Descriptive or leading labels



Why it's wrong

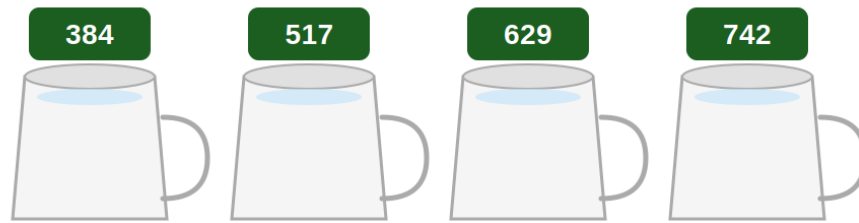
Words like 'Control', 'Test', 'New', or 'Ref' communicate product identity or the purpose of the cup before the panelist has even tasted anything.

Bias introduced

Panelists form expectations based on the label. 'Control' is assumed to be the baseline; 'New' triggers novelty bias. Responses reflect label-driven assumptions, not sensory perception.

Correct labeling approach

Correct example



Requirements

- All four codes are unique 3-digit numbers
- Codes are randomly generated — no sequential or alphabetical pattern
- Codes give no information about which sample is in the cup
- Code assignment is randomized across panelists using a valid tetrad presentation order (e.g., AABB, ABBA, ABAB, BABA, BBAA, BAAB)

Note: The panelist's task is to identify which two of the four cups are alike. Any labeling that hints at pairing, identity, or order reduces the test to guesswork rather than genuine sensory discrimination.